

MyOwnCompiler - Day 07

ProgramNode and Multi-Statement Parsing

Day 7 Objective

Transform the compiler from parsing a single statement into parsing an entire program using ProgramNode.

Theory

ProgramNode is the root AST node that owns all statements in a source file.

```
Program
|
+-- Statement
|
+-- Statement
|
+-- Statement
```

Architecture

```
Source File
  |
  v
Lexer
  |
  v
Tokens
  |
  v
Parser
  |
  v
ProgramNode
```

AST Design

```
ASTNode
  |
  +-- StatementNode
  |       |
  |       +-- PrintStatementNode
  |
  +-- StringLiteralNode
  |
  +-- ProgramNode
```

Parser Walkthrough

```
parse()
while (!isAtEnd())
{
    parseStatement();
}
```

Memory Model

```
vector<unique_ptr<StatementNode>>
```

Complexity

```
Time  O(n)
Space O(n)
```

Interview Questions

1. Why ProgramNode?
2. Why use a root AST node?
3. Why polymorphism?
4. Why `vector<unique_ptr<StatementNode>>`?
5. What is ownership in C++?
6. Why is parsing $O(n)$?

Day 8 Preview

ExpressionNode
NumberLiteralNode

```
print(10);  
print(25);
```